NIELIT Bhubaneswar

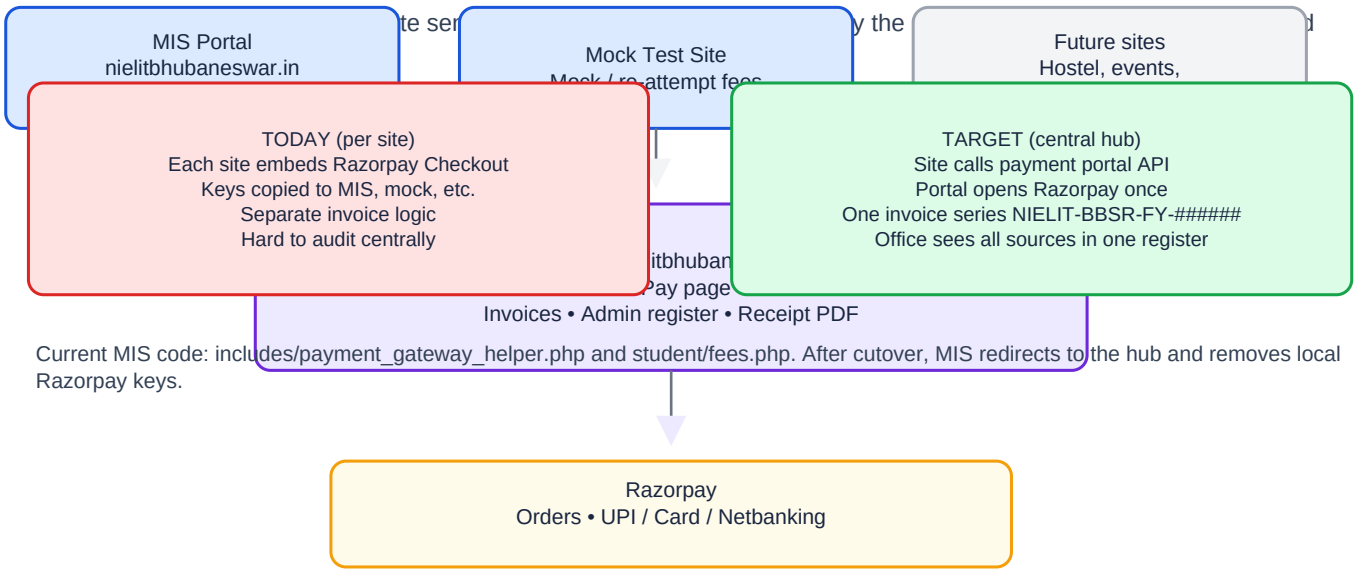
Central Payment Portal

Architecture, Integration & Implementation Report

Portal URL: https://payment.nielitbhubaneswar.in
Gateway: Razorpay (single integration point)
Date: 29 October 2026

- One portal for MIS, mock test, and future sites
- Central invoices and payment register
- Websites never call Razorpay directly
- Signed server-to-server checkout API
- Webhook-confirmed payments only

Diagram 1 — Centralized architecture



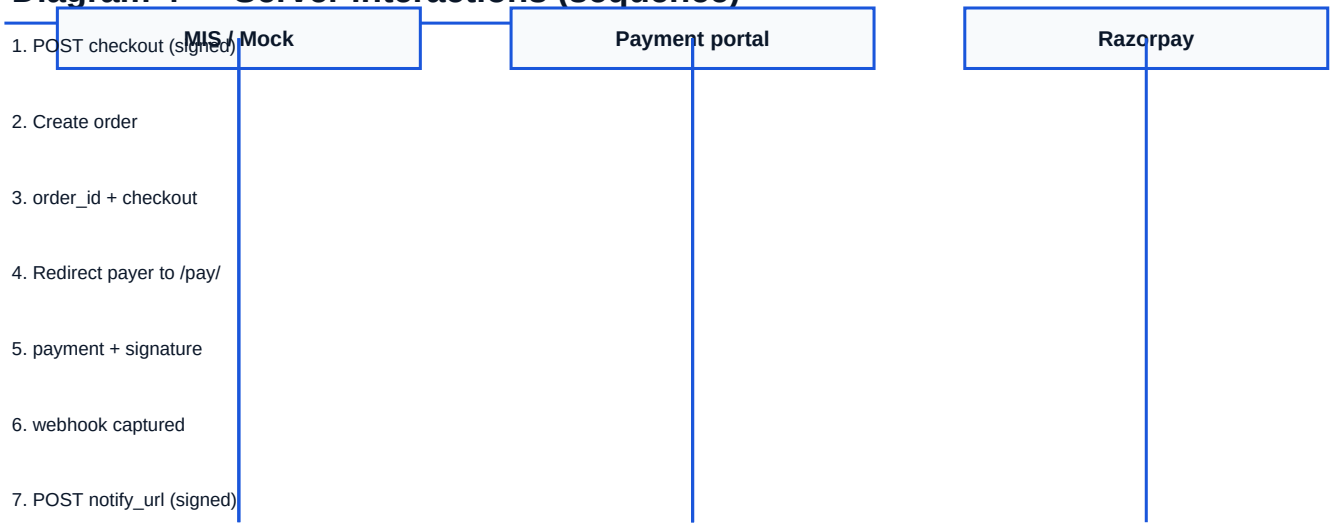
Current MIS code: includes/payment_gateway_helper.php and student/fees.php. After cutover, MIS redirects to the hub and removes local Razorpay keys.

Razorpay Key Secret stays ONLY on the payment server — not on MIS or mock test.

Diagram 2 — Before vs after

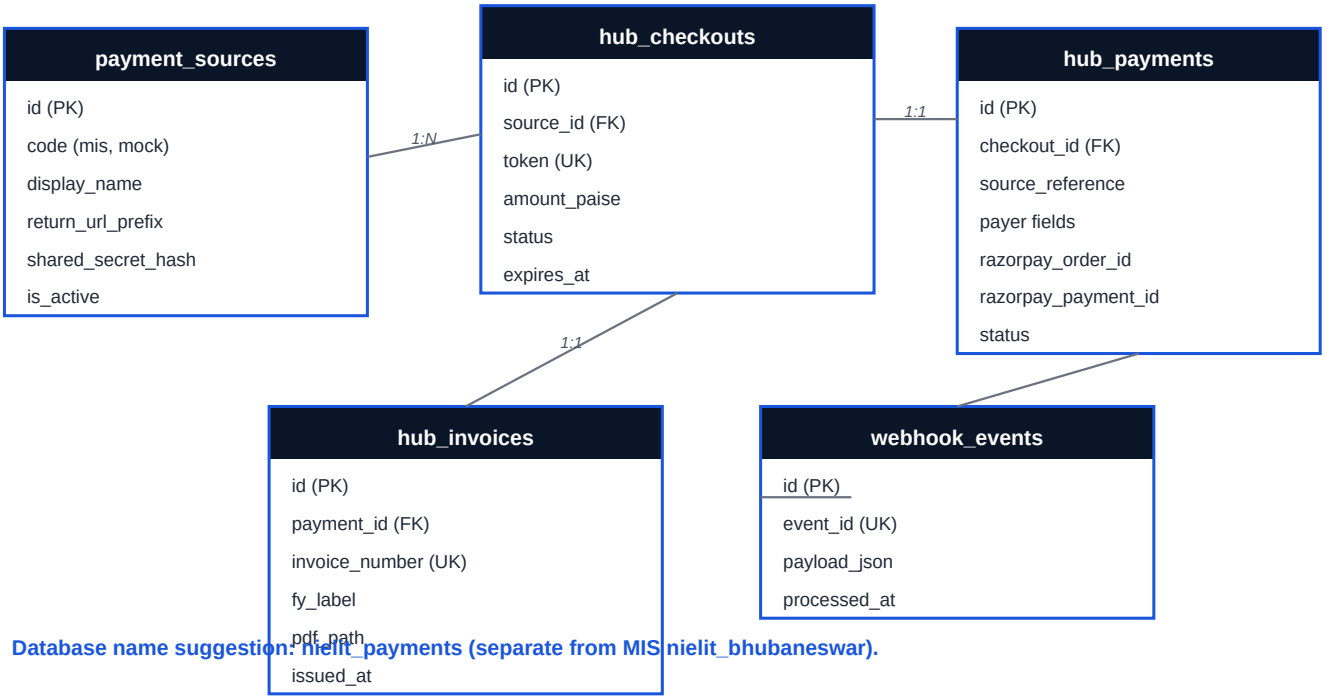
Diagram 3 — End-to-end payment flow

Diagram 4 — Server interactions (sequence)



The source website marks a fee as paid only after step 7 (signed notify) or after querying GET /api/v1/payments/{id} with status paid.

Diagram 5 — Payment portal database (ER)



Database name suggestion: nielit_payments (separate from MIS nielit_bhubaneswar).

Diagram 6 — Security layers

Layer 1: HTTPS only on payment.nielitbhubaneswar.in

Layer 2: HMAC signature on checkout request (X-Payment-Signature)

Layer 3: Return URL whitelist per registered source

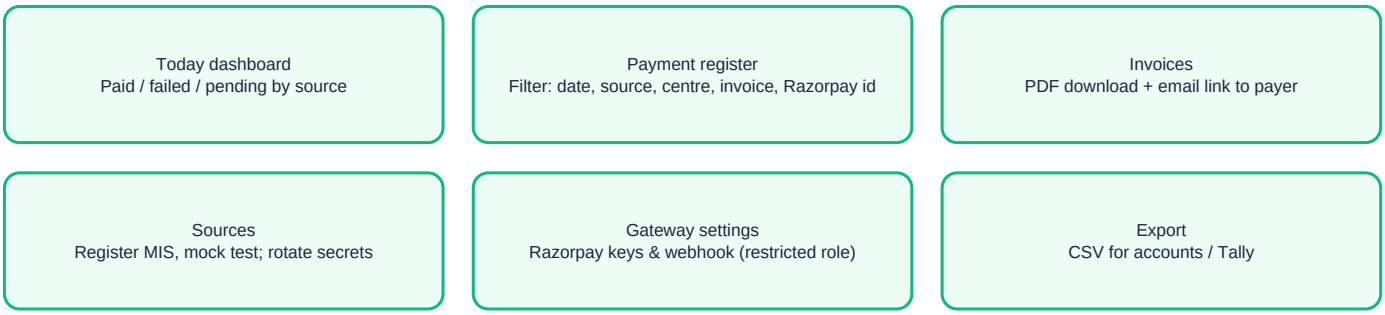
Layer 4: One-time checkout token with expiry (e.g. 30 min)

Layer 5: Razorpay return signature verification

Layer 6: Webhook secret + idempotent event_id storage

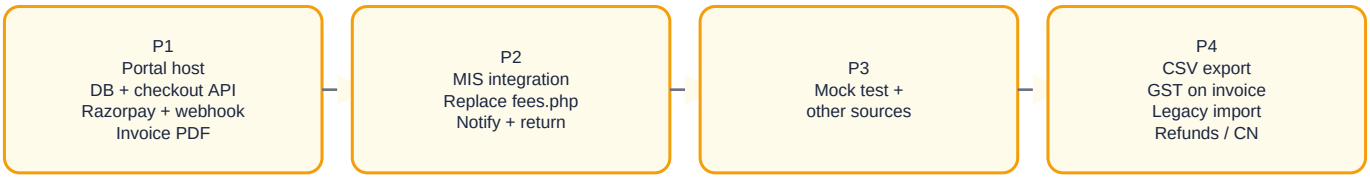
Layer 7: Signed notify to MIS / mock before local “paid” flag

Diagram 7 — Admin portal modules



Answers for accounts: Where was the transaction? ? source website + return host. To whom? ? payer name, email, phone, external student id from that site.

Diagram 8 — Implementation phases



Do not disable current MIS Razorpay until Phase 2 completes one live payment.

1. Executive summary

NIELIT Bhubaneswar can operate one payment hub at payment.nielitbhubaneswar.in. MIS, the mock test website, and any future property redirect payers there. The hub integrates with Razorpay once, stores every transaction, issues a single invoice number series, and notifies each source site when money is captured. This is technically feasible with a signed REST API and is aligned with how the MIS already stores online_course_payments today.

2. Connected websites

Source code	Website	Payment examples
mis	nielitbhubaneswar.in	Course fee, workshop fee
mock	Mock test portal	Mock test fee, re-attempt
(future)	New NIELIT sites	Hostel, recruitment, events

3. Checkout API (summary)

POST https://payment.nielitbhubaneswar.in/api/v1/checkout with signed JSON body. Required fields: source, reference, amount (paise), currency, purpose, description, payer {name, email, phone, external_id}, context {centre, course_id, ...}, return_url, notify_url.

Field	Example
source	mis
reference	MIS-FEE-18422
amount	50000 (= Rs 500.00)
return_url	https://nielitbhubaneswar.in/student/payment_return.php
notify_url	https://nielitbhubaneswar.in/api/payment_hub_notify.php

4. Checkout and payment status

Status	Meaning
created	Payment link issued
attempted	Razorpay order created
paid	Captured; invoice issued
failed	Razorpay failure
cancelled	User closed checkout
expired	Link past allowed time

5. Invoice series

Format: NIELIT-BBSR-{financial year}-{6 digits}. Example: NIELIT-BBSR-2026-27-000184. Financial year April–March (same as Report Monitor). Credit notes: NIELIT-BBSR-CN-2026-27-000012 linked to original invoice.

6. MIS changes after go-live

#	Change
1	student/fees.php redirects to hub; no checkout.razorpay.com on MIS
2	Remove Razorpay secrets from MIS admin (or show hub-only message)
3	online_course_payments + hub_payment_id, hub_invoice_number
4	api/payment_hub_notify.php verifies signed paid callback
5	payment_return.php polls hub status before unlock
6	Optional legacy import of old paid rows into hub register

7. Razorpay configuration

Setting	Value
Webhook URL	https://payment.nielitbhubaneswar.in/api/v1/razorpay/webhook
Events	payment.captured, payment.failed, refund.processed
Order notes	portal_payment_id, source, source_reference
Keys storage	Payment portal server only

8. Hosting

Item	Recommendation
DNS	payment.nielitbhubaneswar.in ? dedicated app
Database	MySQL nielit_payments
PHP	Same major version as production MIS
Secrets	Environment variables / outside web root
Backup	Daily DB + invoice PDF storage

9. Decisions before build

#	Question for accounts / IT
1	One Razorpay merchant for all sites, or per centre?
2	Invoice prefix final text and GST on PDF?
3	Who gets payment admin login?
4	First live integration after MIS: mock test or other?

10. Related document

Full markdown specification: docs/payment-portal/CENTRAL_PAYMENT_PORTAL.md. This PDF is generated by docs/generate_central_payment_portal_pdf.php.